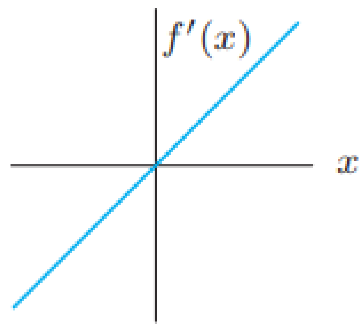
1. (10 pts) Use the following figure about $f'(x)$ to answer the questions:
   (a) Over what intervals is f increasing? Decreasing?
   (b) Does f have local maximum or local minimum? If so, which and where?



2. (10 pts)
   (a) Determine whether the following limit exists.

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}.$$

   (b) Is the function continuous along $y = x$?

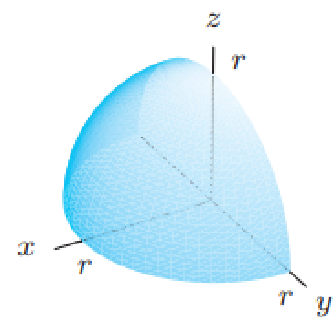
$$h(x,y) = \begin{cases} x^2 & y \geq x, \\ -1 & y < x. \end{cases}$$

3. (10 pts) A plane has equation $z = 5x - 2y + 7$.
   (a) Find a value of λ making the vector $(\lambda, 1, 0.5)$ normal to the plane.
   (b) Find a value of a so that the point $(a+1, a, a-1)$ lies on the plane.

4. (10 pts) Find the equation of the tangent plane of $z = e^y + x + x^2 + 6$ at the point $(1, 0, 9)$

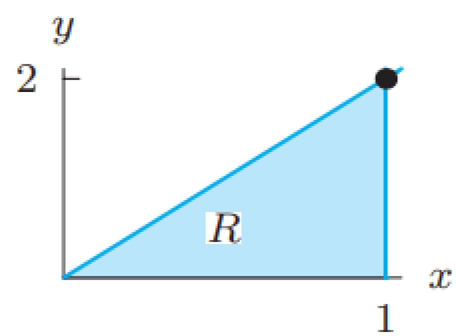
5. (10 pts)
   (a) What is the rate of change of $f(x,y) = 3xy + y^2$ at the point $(2,3)$ in the direction $(3,-1)$?
   (b) What is the direction of maximum rate of change of f at $(2,3)$?
   (c) What is the maximum rate of change?

6. (10 pts) We want to express the volume of the quarter sphere below as an integral of the spherical coordinate $\int_W f\, dV$ where W is the quarter sphere and $dV = \rho^2 \sin(\phi)\, d\rho\, d\phi\, d\theta$.



   (a) (2 pts) What must the function f be?
   (b) (5 pts) Write the limits of the integration.
   (c) (3 pts) Evaluate the integral.

7. (10 pts) Integrate the function $f(x, y) = x + y$ over the shaded region below.



8. (10 pts) Let $F(x) = \int_0^x \sin(\pi t) dt$.

   (a) Evaluate $F(1)$.
   (b) For what values of x is $F(x)$ positive? negative?

9. (10 pts) Find the volume of the solid that lies under the paraboloid $z = 4 - x^2 - y^2$ and above the xy-plane.

10. (10 pts) Decide whether the following statements are true or false.

   (a) If all partial derivatives of $f(x, y)$ exist at a point (a, b), then f must be continuous at (a, b).
   (b) The iterated integral $\int_0^2 \int_0^{x^2} f(x, y)\, dy\, dx$ is equal to $\int_0^4 \int_{\sqrt{y}}^2 f(x, y)\, dx\, dy$.
   (c) If $f(x, y)$ is a continuous function on a closed region R and $f(x, y) \geq 0$ for all (x, y) in R, then $\iint_R f(x, y)\, dA = 0$ implies that $f(x, y) = 0$ at all points in R.

(d) For any continuous function $f(x, y)$, the double integral $\iint_R f(x, y)\, dA$ represents the volume of the solid trapped between the surface $z = f(x, y)$ and the xy-plane over the region R.

(e) If $\nabla f(a, b) = \mathbf{0}$, then f must have a local maximum or a local minimum at (a, b).